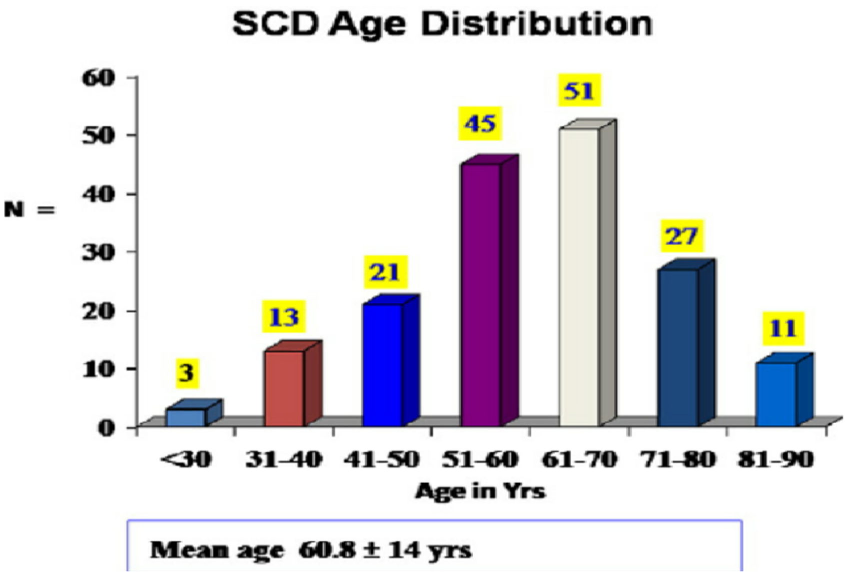
Cardiovascular Risk Classification and Treatment Recommendation System

A Unified Machine Learning Pipeline for Clinical Decision Support

1. Problem Statement

Cardiovascular disease is one of the leading causes of death, especially among the elderly. A major clinical challenge is that many aging individuals who maintained healthy lifestyles in their youth may still have hidden, serious cardiovascular risk due to the natural aging process of their hearts and blood vessels. As these patients often appear to be generally healthy, sudden and fatal cardiac events such as cardiogenic shock or sudden heart failure may occur unexpectedly with rapid fluctuations in vital signs.



The distribution of the cohort of sudden cardiac deaths by age is depicted. The peak distribution is seen in the age group of 50 – 70 years.

`https://www.researchgate.net/figure/The-distribution-of-the-cohort-of-sudden-cardiac-deaths-by-age-is-depicted-The-peak_fig2_47299203`

This project addresses both gaps in a unified ML pipeline: automatically classifying a patient's cardiovascular risk from clinical vital signs, then recommending a targeted treatment intervention and estimating survival probability under that treatment using a biologically grounded scoring formula.

2. Summary

2.1 Risk Classification

Two datasets are combined to enrich the feature space (see Section 3). Engineered clinical features (age bins, blood pressure risk levels, and the Heart Rate Stress Index) are fed into a Random Forest and an XGBoost classifier. The output is a cardiovascular risk category (**Low** / **Medium** / **High**) with a confidence score.

2.2 Treatment Recommendation and Survival Prediction

The Part 1 output is merged with extended patient metadata from MIMIC-IV and passed to a second XGBoost model that recommends an intervention type and intensity. For high-risk patients flagged for cardiac radioablation (Stereotactic Body Radiation Therapy, SBRT), the recommended dose is biologically validated through the Biologically Effective Dose (BED) formula, which is the same physics-based equation used in clinical oncology:

$$\mathrm{BED} = D\left(1 + \frac{d}{\alpha/\beta}\right) \tag{1}$$

where D is the total dose (Gy), d is the dose per fraction, and $\alpha/\beta = 10\,\mathrm{Gy}$ for cardiac tissue.

For lower-risk patients, medication intensity is calibrated via the GRACE Risk Score. A survival probability estimate is then produced, yielding a counterfactual output of the form: *"Without intervention: 58% two-year survival. With recommended SBRT at X Gy: 81% two-year survival."*

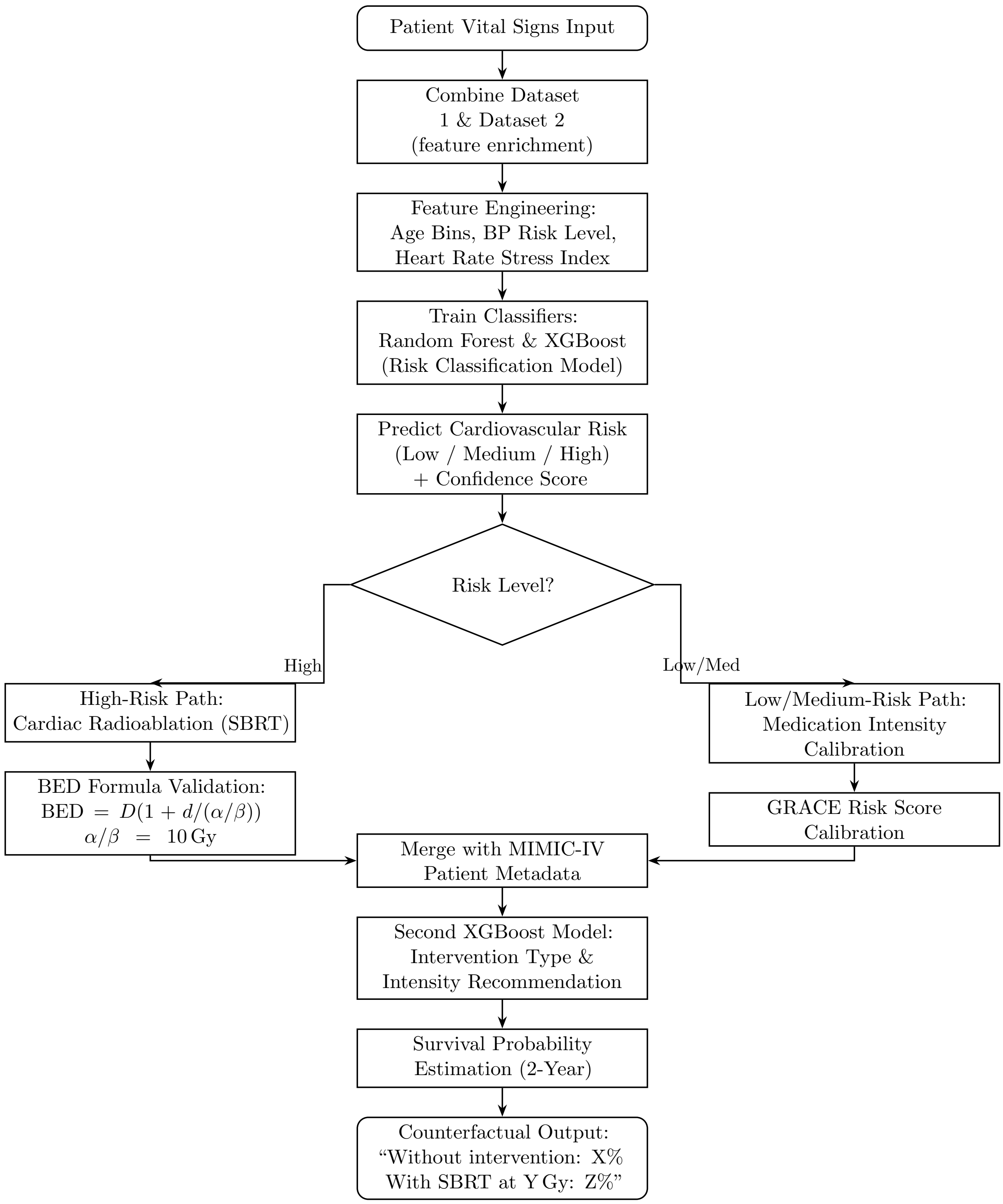
3. Datasets

Four data sources are combined across the two parts of the pipeline. Part 1 uses two public clinical datasets that are harmonised and merged to enrich the training set; Part 2 draws on the MIMIC-IV electronic health record database and a small literature-derived dose-record table used for BED calibration.

Dataset	**Source**	**Format**	**Usage**
Heart Failure Prediction	Kaggle (fedesoriano)	CSV, 918 rows, 11 features	Part 1: primary classification labels and features
Cleveland Heart Disease (UCI)	UCI ML Repository	CSV, 303 rows, 14 features	Part 1: supplementary set merged to add angiographic features (*ca*, *thal*) and increase training sample size
MIMIC-IV Clinical	PhysioNet	CSV / relational	Part 2: treatment history, medication records, survival outcomes
SBRT/VT Dose Records	Literature-derived	CSV	BED formula calibration for the high-risk radiation path

Why add the UCI Cleveland dataset? The Kaggle Heart Failure dataset contains binary labels but lacks detailed angiographic features. The UCI Cleveland set adds clinically significant predictors such as *ca* (number of major vessels coloured by fluoroscopy) and *thal* (thalassemia type). Merging both datasets after harmonising column names and encoding schemes produces a richer, larger training set for Part 1, and provides a natural opportunity to demonstrate data integration skills during the preprocessing stage.

4. System Flowchart



End-to-end system flowchart of the cardiovascular ML pipeline.

5. Example Cases for Running the Model

The following walk-throughs trace four representative patients end-to-end through the pipeline, illustrating every possible branch: Low risk, Medium risk, High risk without arrhythmia, and High risk with arrhythmia (the SBRT path). Each case shows the input vital signs, the Part 1 classification, the Part 2 recommendation, and the counterfactual survival output.

Case A Low-Risk Patient

Input vitals: 42-year-old, resting heart rate 68 bpm, blood pressure 118/76 mmHg, cholesterol 185 mg/dL, fasting blood sugar normal, no exercise-induced angina, normal resting ECG.

Part 1 output: Risk Category = **Low**, confidence = 91%.

Part 2 path: Routed to medication-intensity calibration. GRACE Risk Score = 78 (low band). Recommendation: lifestyle optimisation plus low-dose statin monotherapy; no antiplatelet indicated.

Counterfactual output: *"Without intervention: 95% two-year survival. With recommended low-dose statin: 97% two-year survival."*

Case B Medium-Risk Patient

Input vitals: 58-year-old, resting heart rate 84 bpm, blood pressure 142/88 mmHg, cholesterol 232 mg/dL, fasting blood sugar mildly elevated, occasional chest discomfort on exertion, ST-T abnormality on resting ECG.

Part 1 output: Risk Category = **Medium**, confidence = 83%.

Part 2 path: Routed to medication-intensity calibration. GRACE Risk Score = 132 (intermediate band). Recommendation: dual therapy with a beta-blocker and a moderate-intensity statin, plus low-dose aspirin; follow-up stress echocardiogram scheduled.

Counterfactual output: *"Without intervention: 81% two-year survival. With recommended dual regimen: 92% two-year survival."*

Case C High-Risk Patient without Arrhythmia

Input vitals: 67-year-old, resting heart rate 96 bpm, blood pressure 162/98 mmHg, cholesterol 268 mg/dL, fasting blood sugar elevated, exercise-induced angina present, ST depression on resting ECG, no documented ventricular tachycardia.

Part 1 output: Risk Category = **High**, confidence = 88%.

Part 2 path: Decision node checks for arrhythmia. Because no ventricular tachycardia is documented, the SBRT branch is not triggered. Routed instead to aggressive medication regimen: high-intensity statin, beta-blocker, ACE inhibitor, dual antiplatelet therapy, and urgent referral for coronary angiography.

Counterfactual output: *"Without intervention: 62% two-year survival. With recommended aggressive medical therapy: 84% two-year survival."*

Case D High-Risk Patient with Arrhythmia (SBRT Path)

Input vitals: 71-year-old, resting heart rate 108 bpm, blood pressure 158/94 mmHg, cholesterol 245 mg/dL, prior myocardial infarction, recurrent monomorphic ventricular tachycardia refractory to catheter ablation and antiarrhythmic drugs.

Part 1 output: Risk Category = **High**, confidence = 94%.

Part 2 path: Decision node identifies refractory ventricular tachycardia. The pipeline routes the patient onto the cardiac SBRT branch. The recommended total dose is $D = 25\,\text{Gy}$ in a single fraction ($d = 25\,\text{Gy}$). BED validation:

$$\text{BED} = 25\left(1 + \frac{25}{10}\right) = 87.5\,\text{Gy}.$$

This sits inside the published clinical window for cardiac radioablation, so the dose is accepted.

Counterfactual output: *"Without intervention: 58% two-year survival. With recommended SBRT at 25 Gy: 81% two-year survival."*

Coverage Summary

Case	Part 1 Output	Part 2 Branch	Recommendation
A	Low	Medication (GRACE low)	Lifestyle + low-dose statin
B	Medium	Medication (GRACE mid)	Beta-blocker + statin + aspirin
C	High	Medication (no arrhythmia)	Aggressive multi-drug regimen
D	High	SBRT (arrhythmia present)	25 Gy single-fraction radioablation

6. Similar Projects & References

The following works and resources are directly related to the methodology and clinical context of this project. All links were verified as publicly accessible.

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